Librarians Critical Digital Literacy Guide to Smart Software Selections

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We would like to generate a transformative shift in the way academic libraries and their parent institutions address digital tools. Prevailing attitudes and current administrative practices do not treat digital tools as part of library collections, which puts digital humanities (DH) researchers at a disadvantage compared to traditional research. A shift is required for software to be recognized as an integral part of the collection (parallel to books, films, journals, etc.). Barring such a shift, we can compensate somewhat by providing instruction in critical digital literacy toward finding and selecting software. Librarians can support DH researchers gaining the understanding, disposition, and aptitudes for searching, evaluating, and selecting tools that adequately serve their requirements. These skills foster equity and open scholarship in the research lifecycle.

This presentation introduces a scaffolded pedagogical guide to teach researchers the practice of discovering, understanding, and evaluating DH tools and techniques that align with research objectives. The guide, inspired by a larger research initiative, is scheduled to appear in a forthcoming publication of the Association of College and Research Libraries (ACRL) Digital Humanities Cookbook. The guide acknowledges challenges for assessing needs and selecting the appropriate DH tools while planning a project. Activities in the guide help researchers plan and reflect on their approaches and discovery of digital tools. It presents exercises that develop researchers' critical digital literacy through evaluating the strengths and limitations of various technologies based on well-defined criteria. With practice, researchers build confidence and the knowledge needed to navigate the technological landscape.

Before designing the pedagogical guide, the principle investigators (PIs) initiated a 2023 study of North American academic institutions to address a gap in academic library selection practices, namely an absence of policies governing the acquisition, use, and support of software that impacts DH research initiatives. Publications at least as far back as 1995 refer to the examination of software but do not fully explore its place within collection development (McDougall and Squires 1995). Academic libraries follow their developed selection criteria as "...an administrative device designed to facilitate implementation of decisions about materials... (Gorman and Howes 1989)" that support community needs as well as advance research and teaching requirements. These policies culminate from years of established expertise and documented literature that exemplify library best practices (Futas 1995). To conduct the study, we implemented a two-stage approach: first a web-based survey, followed by semi-structured interviews. The PIs based in Canadian institutions, surveyed both the Canadian academic library landscape and for breadth, the greater quantity of American universities. The PIs identified institutions' memberships to the Digital Library Federation as an indicator of institutional support for digital research. The survey was distributed to 236 universities, resulting in 49 responses, of which 12 participants from different institutions agreed to be interviewed.

Preliminary analysis of the research results revealed that many academic libraries do not follow a formal selection process for software. Trends among libraries point to a reactive, ad-hoc selection method rather than a proactive and systematic process. Ad hoc methods may address immediate concerns or requests but don't build long-term strategic understanding of academic and community needs. They may result in libraries over- or under-supporting software tools without adequately supporting researchers' needs. Unformalized selection practices could inadvertently create inequities and run the risk of being shaped by personal biases that address the preferences of small groups rather than the varied needs of the broader user community. Unable to rely on library collection work the way researchers might with traditional resources, they face challenges of discovering and evaluating anew the tools for their projects. This emphasizes researchers' needs for critical digital literacy skills.

The American Library Association (ALA) developed a selection policy toolkit (American Library Association 2017) for print and digital resources with fundamental guiding principles that consider cost, relevance, ease of use, presentation of diverse perspectives, objectivity, and resources' long term significance but it does not specifically target software selection. The toolkit aligns with the Association of College and Research Libraries (ACRL) Framework for Information Liter-

acy for Higher Education, which reinforces an understanding of openness, access and accessibility, and critical thinking and evaluation (Association of College and Research Libraries 2015). Further discourse on digital literacy skills comes from The Canadian Association of Research Libraries (CARL) Digital Literacy Framework (DLF). Both frameworks call attention to search processes. The CARL DLF calls for the ability to "evaluate and choose devices, applications, software, and systems relevant to different search tasks." (Digital Literacy Task Group of the Advancing Teaching and Learning Committee 2024) and identifies the ability to use digital technologies and techniques in creating new knowledge (Digital Literacy Task Group of the Advancing Teaching and Learning Committee 2024). This implies that researchers must consider the context in which new information will exist, and searching must work in service of that end. Context includes things such as the format other people will encounter it in, the value that it will provide and who will be capable of accessing that value, and implications for its future availability. The framework's scope however does not reach a depth necessary to address specific needs of DH researchers.

In our pedagogical guide, we walk researchers through elements of the selection process. The guide breaks the process into three stages: remembering and understanding; applying and analyzing; and evaluating. The first stage introduces DH terminology and highlights concepts like "interdisciplinarity", "collaboration", "technological impact", and "intentional changes to the research process," to raise awareness of different aspects of software that enable or hinder research. The second stage trains participants to compare DH tools in support of different methodologies. Participants conduct searches, first articulating their needs, scope of search systems, and context for what they are searching, in order to discover and reflect on the benefits and drawbacks of each technological solution. They examine requirements and functionality, and supports to determine the technology's suitability for meeting research goals. Finally, the guide encourages participants to consider ethical motivations and responsibilities in employing DH techniques. It emphasizes understanding data privacy and sovereignty, access, reproducibility, preservation, respect for subject and historical context, an awareness of cultural sensitivities, and user experience.

We hope that DH practitioners and supporters will leave our presentation with an awareness of the critical digital literacy skills necessary for research software selection as well as concrete pedagogical techniques to that end.

Bibliography

American Library Association (2017): "Selection Criteria", in: Selection & Reconsideration Policy Toolkit for Public, School, & Academic Libraries. https://www.ala.org/tools/challengesupport/selectionpolicytoolkit/criteria.

Association of College and Research Libraries (2015): "Framework for Information Literacy for Higher Education" American Library Association. http://www.ala.org/acrl/standards/ilframework.

Digital Literacy Task Group of the Advancing Teaching and Learning Committee (2024): Digital Literacy Framework. Canadian Association of Research Libraries.: https://www.carl-abrc.ca/advance-teaching-learning/digital-literacy/.

Futas, Elizabeth (1995): *Collection Development Policies and Procedures*. 3rd ed. A Neal-Schuman Professional Book. Phoenix, Ariz.: Oryx Press.

Gorman, G. E. / Howes, Brian R. (1989): Collection Development for Libraries. *Topics in Library and Information Studies*. London : Bowker-Saur. http://www.gbv.de/dms/hbz/toc/ht004735895.PDF.

McDougall, Anne / Squires, David (1995): "A Critical Examination of the Checklist Approach in Software Selection", in: *Journal of Educational Computing Research* 12, 3: 263–74. https://doi.org/10.2190/8X8Y-2URM-FG7E-5LB3.